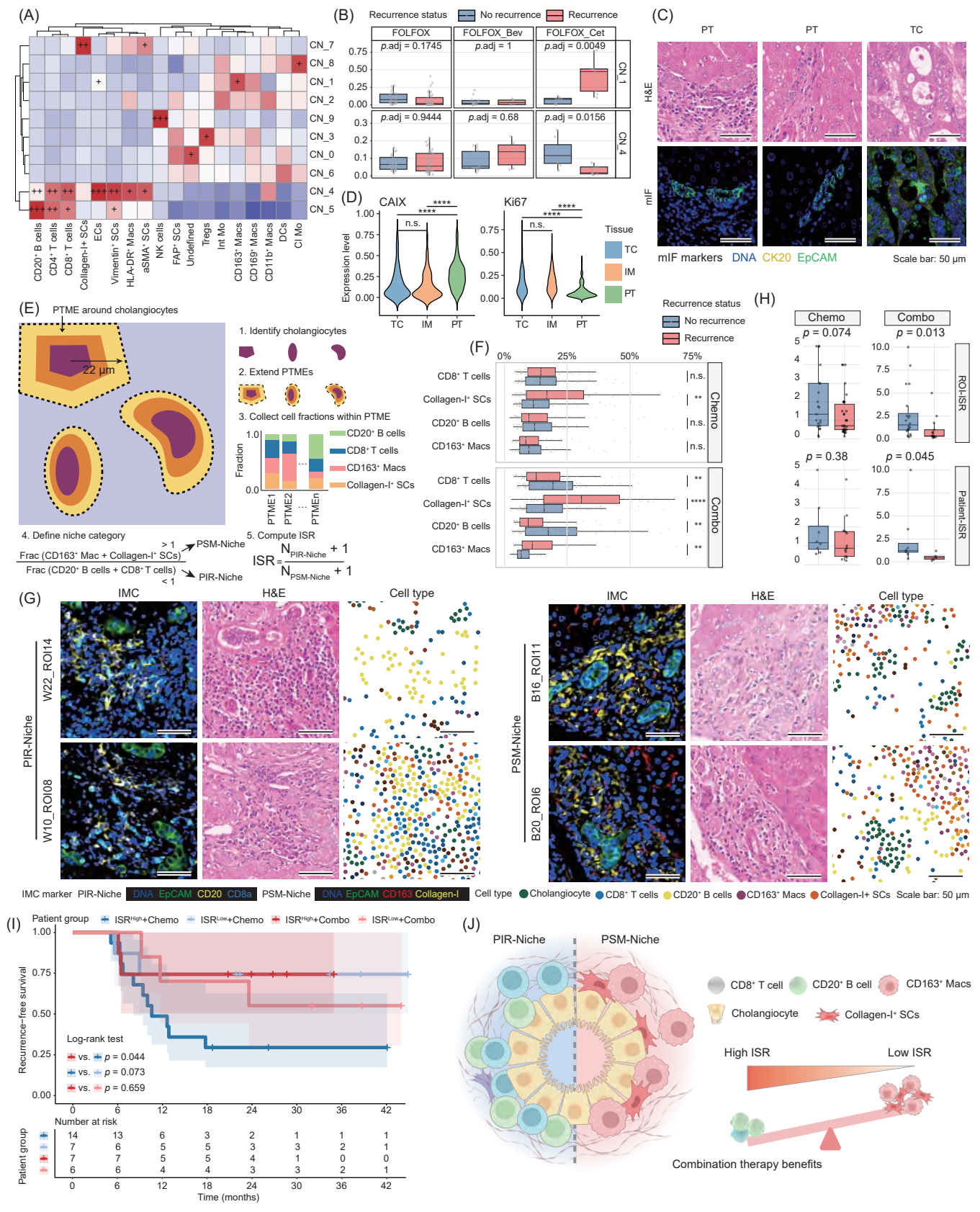

(A)
CN_7
CN_8
CN_1
CN_2
CN_9
CN_3
CN_0
CN_6
CN_4
CN_5
CD20+ B cells
CD4+ T cells
CD8+ T cells
Collagen-I+ SCs
ECs
Vimentin+ SCs
HLA-DR+ Macs
aSMA+ SCs
NK cells
FAP+ SCs
Undefined
Tregs
Int Mo
CD163+ Macs
CD169+ Macs
CD11b+ Macs
DCs
Cl Mo
(B)
Recurrence status
No recurrence
Recurrence
FOLFOX
FOLFOX_Bev
FOLFOX_Cet
p.adj = 0.1745
p.adj = 1
p.adj = 0.0049
p.adj = 0.9444
p.adj = 0.68
p.adj = 0.0156
CN_1
CN_4
(C)
PT
PT
TC
H&E
mIF
mIF markers DNA CK20 EpCAM
Scale bar: 50 μm
(D)
CAIX
Ki67
n.s.
****
Expression level
TC
IM
PT
Tissue
(E)
PTME around cholangiocytes
22 μm
1. Identify cholangiocytes
2. Extend PTMEs
3. Collect cell fractions within PTME
Fraction
CD20+ B cells
CD8+ T cells
CD163+ Macs
Collagen-I+ SCs
PTME1
PTME2
PTMEn
4. Define niche category
Frac (CD163+ Mac + Collagen-I+ SCs) / Frac (CD20+ B cells + CD8+ T cells)
> 1
PSM-Niche
< 1
PIR-Niche
5. Compute ISR
ISR = (N_PIR-Niche + 1) / (N_PSM-Niche + 1)
(F)
0%
25%
50%
75%
CD8+ T cells
Collagen-I+ SCs
CD20+ B cells
CD163+ Macs
n.s.
**
****
Chemo
Combo
(H)
Recurrence status
No recurrence
Recurrence
Chemo
Combo
p = 0.074
p = 0.013
p = 0.38
p = 0.045
ROI-ISR
Patient-ISR
(G)
IMC
H&E
Cell type
PIR-Niche
PSM-Niche
W22_ROI14
W10_ROI08
B16_ROI11
B20_ROI6
IMC marker PIR-Niche DNA EpCAM CD20 CD8a PSM-Niche DNA EpCAM CD163 Collagen-I
Cell type Cholangiocyte CD8+ T cells CD20+ B cells CD163+ Macs Collagen-I+ SCs
Scale bar: 50 μm
(I)
Patient group
ISRHigh+Chemo
ISRLow+Chemo
ISRHigh+Combo
ISRLow+Combo
Recurrence−free survival
Log-rank test
vs. p = 0.044
vs. p = 0.073
vs. p = 0.659
Number at risk
Patient group
Time (months)
(J)
PIR-Niche
PSM-Niche
CD8+ T cell
CD20+ B cell
CD163+ Macs
Cholangiocyte
Collagen-I+ SCs
High ISR
Low ISR
Combination therapy benefits